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Deep Learning Project Report on

"CROP CARE: DEEP LEARNING BASED PLANT DISEASE DETECTION SYSTEM"

Submitted in Partial fulfilment of the Requirements for the Degree of

Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence & Machine Learning)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)

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CERTIFICATE

This is to certify that the Assignment report work entitled **"Crop Care: Deep Learning Based Plant Disease Detection System"** has been carried out by **Akshita N Nair (1CR22CI003)** and **Harshini R (1CR22CI020)**, Bonafide students of CMR Institute of Technology, Bengaluru in partial fulfilment for the award of the Degree of **Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning)** of the Visvesvaraya Technological University, Belagavi during the year **2025-2026.** It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report deposited in the departmental library. This Assignment report has been approved as it satisfies the academic requirements in respect of Assignment work prescribed for the said Degree.

Signature of	**Signature of**
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DECLARATION

Ms. **Akshita N Nair (1CR22CI003)** and Ms. **Harshini R (1CR22CI020)** hereby declare that the Assignment report entitled "**Crop Care: Deep Learning Based Plant Disease Detection System**" has been carried out by me under the guidance of **Prof. Navya VK**, **Assistant Professor**, **Department of AI&ML**, CMR Institute of Technology, Bengaluru, in partial fulfilment of the requirement for the degree of BACHELOR OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING), of Visvesvaraya Technological University, Belgaum during the academic year 2025-2026. The work done in this Assignment report is original and it has not been submitted for any other degree in any university.

Place: Bangalore

Date: 01/11/2025

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ABSTRACT

Early detection of plant diseases plays a crucial role in improving agricultural productivity and minimizing economic losses. Traditional diagnostic methods rely heavily on expert observation and laboratory analysis, which are often time-consuming, costly, and inaccessible to small-scale farmers. In this context, the proposed Crop Care system leverages the power of deep learning to automatically identify plant diseases from leaf images, enabling rapid and reliable diagnosis through intelligent image classification.

The core of this project is a Convolutional Neural Network (CNN) model trained to classify crop leaf images into multiple disease categories across several plant types. The model architecture was designed to progressively extract low- to high-level visual features using stacked convolution and pooling layers, followed by dense layers for classification.

Techniques such as data augmentation, dropout regularization, and Adam optimization were employed to improve generalization and minimize overfitting. The model was trained and validated on a diverse dataset of 15 disease classes, achieving high accuracy and robust performance across crops like tomato, potato, maize, and pepper.

For real-world usability, the trained model was converted to TensorFlow Lite (TFLite) format for efficient deployment on mobile devices, supporting on-device inference without internet connectivity. This integration demonstrates the potential of deploying deep learning solutions beyond laboratory environments, directly benefiting the agricultural community. Overall, the project showcases a complete deep learning pipeline—from data preprocessing to model compression and deployment—highlighting the transformative impact of deep learning in precision agriculture.

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TABLE OF CONTENTS

S. No.	Contents	Page No.
1	Certificate	**1**
2	Declaration	**2**
3	Abstract	**3**
4	Acknowledgement	**4**
5	Group Members List	**5**
6	Table of Contents	**6**
7	Introduction And Objectives	**7**
8	System Design and Working	**9**
9	Implementation	**13**
10	Results and Conclusion	**15**

1. INTRODUCTION AND OBJECTIVES

Agriculture forms the backbone of many economies, yet it remains one of the sectors most vulnerable to plant diseases, which can severely affect crop yield and food security. Early and accurate detection of these diseases is critical for ensuring timely intervention and preventing large-scale agricultural loss. In recent years, deep learning—a branch of artificial intelligence inspired by the human brain—has emerged as a powerful tool for image-based recognition and classification tasks. Leveraging this technology, the Crop Care system was developed as a deep learning-based solution for automatic plant disease detection, enabling real-time diagnosis from leaf images captured using a smartphone camera.

This section presents the background motivation behind the project, the problem being addressed, the specific objectives of the system, and its overall scope. The focus lies on how deep learning methodologies, particularly Convolutional Neural Networks (CNNs), can effectively learn to identify visual symptoms of plant diseases from image data.

1.1. Background and Motivation

Plant diseases are among the leading causes of crop failure worldwide, reducing yield quality and quantity significantly. Traditional detection methods depend on agricultural experts who visually inspect plants and identify infections based on their experience. However, such approaches are often subjective, time-intensive, and inaccessible to farmers in remote or resource-limited regions. Additionally, environmental factors such as lighting, camera quality, and disease stage can make manual diagnosis more challenging.

With the rapid advancement of computer vision and deep learning, automated image-based disease detection has become a viable alternative. Deep learning models, particularly CNNs, can automatically extract hierarchical patterns from plant images—learning to differentiate between healthy and diseased leaves without manual feature engineering. This not only improves diagnostic accuracy but also enables scalable, real-time disease detection that can operate even on mobile devices.

The motivation behind the Crop Care system lies in bridging the gap between cutting-edge AI research and practical agricultural applications, making intelligent disease detection accessible, affordable, and efficient.

1.2. Problem Definition

The primary problem addressed in this project is the automatic identification of plant diseases from digital images of crop leaves using deep learning. The variability in disease symptoms—such as changes in color, texture, and shape—makes this a challenging computer vision task.

The project aims to design and train a CNN capable of classifying plant leaf images into 15 distinct categories, including both healthy and infected conditions across multiple crop species (tomato, potato, maize, and pepper).

Key challenges include:

- Handling variations in lighting, orientation, and image quality.
- Preventing overfitting due to limited labeled datasets.
- Achieving high inference speed for real-time mobile deployment.
- Addressing these challenges required a careful balance between model complexity, dataset diversity, and computational efficiency.

1.3. Objectives

The objectives of the Crop Care deep learning system are as follows:

- To develop and train a CNN-based deep learning model for accurate classification of crop diseases from leaf images.
- To apply data augmentation and regularization techniques to enhance generalization and prevent overfitting.
- To evaluate the model's performance using appropriate metrics such as accuracy and validation loss, ensuring robust classification across multiple crop types.
- To convert and optimize the trained model into TensorFlow Lite format for efficient mobile deployment.
- To enable real-time disease diagnosis directly on mobile devices, ensuring accessibility even in low-connectivity regions.

Through these objectives, the project demonstrates not just theoretical understanding of CNNs but also practical expertise in building deployable deep learning systems.

2. SYSTEM DESIGN AND WORKING

The Crop Care system follows a structured deep learning workflow, beginning from data collection and preprocessing to model training, evaluation, and mobile deployment. The goal of the system is to accurately classify plant leaf images into different disease categories by learning discriminative visual features through a Convolutional Neural Network (CNN).

This section describes the system's overall design and functionality, covering the dataset preparation, architecture rationale, training strategy, and deployment workflow.

2.1. System Overview

The system is designed as an end-to-end pipeline, where the CNN model acts as the core decision-making engine. The process begins when an image of a plant leaf is captured or uploaded by the user. The image is pre-processed to match the input requirements of the CNN and then passed through the trained model for inference. The model outputs probabilities for each disease class, and the one with the highest confidence is selected as the prediction.

Workflow Summary:

1. Image capture/upload by user.
2. Image preprocessing (resize to 256×256, normalization).
3. Forward pass through CNN model (feature extraction and classification).
4. SoftMax layer outputs probability distribution across 15 classes.
5. The class with the highest probability is displayed with its confidence score.
6. On-device inference is achieved through TensorFlow Lite for real-time diagnosis.

This architecture ensures an efficient and scalable design capable of operating both in laboratory testing and real-world field scenarios.

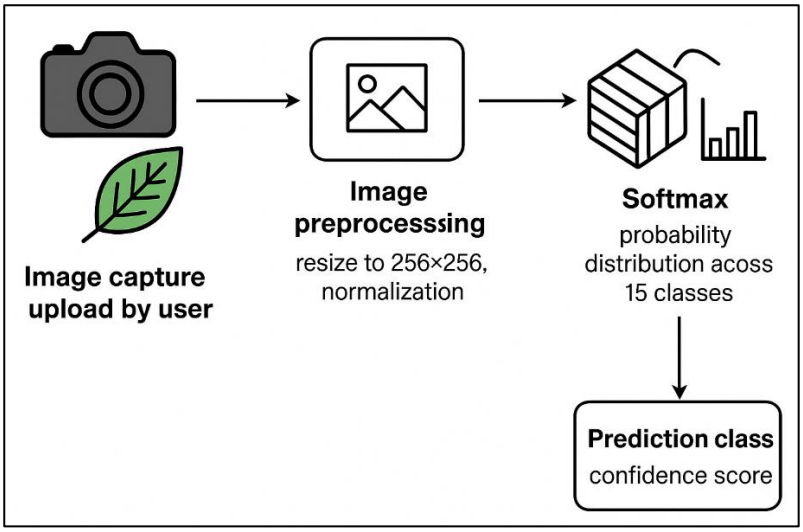


Fig 1. Workflow Summary

2.2. Dataset and Preprocessing

A well-prepared dataset is crucial for effective deep learning model performance. The dataset used in this project consists of RGB images of crop leaves spanning multiple species (tomato, potato, maize, and pepper), each annotated with corresponding disease labels. In total, 15 classes are represented, covering both healthy and diseased samples.

Preprocessing Steps:

- Resizing: All images were resized to 256 × 256 pixels to ensure uniform input dimensions.
- Normalization: Pixel values were rescaled by dividing by 255 to bring them into the [0,1] range, improving numerical stability during training.
- Augmentation: To improve model generalization and reduce overfitting, several augmentation techniques were applied:
    - Shear Transformation (range = 0.2) to simulate geometric distortion.
    - Zoom Transformation (range = 0.2) to create scale variation.
    - Horizontal Flipping to mimic real-world leaf orientations.

  These augmentations effectively increased dataset diversity and made the model robust to environmental variations such as lighting, orientation, and background noise.

Input Specifications:

- Image size: 256 × 256 × 3
- Batch size: 16
- Colour space: RGB
- Class mode: Categorical (one-hot encoding for 15 classes)

The preprocessing pipeline ensured that the model received consistent, high-quality inputs for efficient learning.

2.3. CNN Architecture Design

The backbone of the Crop Care system is a Sequential Convolutional Neural Network (CNN) designed to automatically learn visual features associated with plant diseases. The network architecture consists of four convolutional blocks, each followed by a pooling layer, culminating in dense layers for classification.

Model Architecture:

Input Layer: (256 × 256 × 3)
├─ Conv2D: 32 filters, 3×3 kernel, ReLU activation
├─ MaxPooling2D: 2×2
├─ Conv2D: 64 filters, 3×3 kernel, ReLU activation
├─ MaxPooling2D: 2×2
├─ Conv2D: 128 filters, 3×3 kernel, ReLU activation

├─ *MaxPooling2D: 2×2*
├─ *Conv2D: 128 filters, 3×3 kernel, ReLU activation*
├─ *MaxPooling2D: 2×2*
├─ *Flatten*
├─ *Dense: 512 neurons, ReLU activation*
├─ *Dropout: 0.5*
└─ *Output: 15 neurons, SoftMax activation*

Design Rationale:

- Progressive Feature Extraction: Increasing filters (32→64→128) helps capture simple to complex visual features, from edges to intricate disease textures.
- MaxPooling Layers: Reduce spatial dimensions by half at each stage, providing translation invariance and lowering computation.
- Dropout Regularization: A 50% dropout rate prevents overfitting by deactivating random neurons during training.
- SoftMax Output: Produces a normalized probability distribution across all 15 classes, suitable for multi-class classification.

This CNN structure offers a strong balance between computational efficiency and classification accuracy, making it ideal for deployment on constrained mobile hardware.

2.4. Training Configuration

Model training was performed using TensorFlow 2.x with the Keras API, leveraging GPU acceleration for efficient computation.

Training Parameters:

- *Optimizer:* Adam (Adaptive Moment Estimation)
    - Combines advantages of AdaGrad and RMSProp.
    - Provides adaptive learning rates for faster convergence.
- *Loss Function:* Categorical Cross-Entropy - Measures divergence between predicted and true class distributions.
- *Epochs:* Maximum of 50 (early stopping applied).
- *Batch Size:* 16
- *Validation Split:* Portion of dataset reserved for validation to monitor overfitting.

Callbacks Used:

- *EarlyStopping*: Stops training once validation accuracy stops improving.
- *ModelCheckpoint:* Saves the best-performing model based on validation metrics.
- *ReduceLROnPlateau:* Reduces learning rate when progress stalls.

During training, the model's accuracy and loss curves were plotted for both training and validation datasets, allowing visual monitoring of convergence behaviour and generalization performance.

2.5. Model Conversion and Deployment

Once trained, the Keras model was converted to TensorFlow Lite (TFLite) format to enable mobile inference. This was done using:

```
converter = tf.lite.TFLiteConverter.from_keras_model(model)
tflite_model = converter.convert()
```

Why TensorFlow Lite?

- Optimized for mobile and edge devices.
- Model size reduced by 2–4× through quantization (float32 → int8).
- Faster inference and lower power consumption.
- Enables offline inference — critical for agricultural regions with limited connectivity.

Quantization Impact:

- Slight accuracy trade-off (<1%).
- Significant performance gains in speed and efficiency.
- Lower memory usage, allowing smooth mobile deployment.

While the model is integrated into a lightweight mobile app (built using Flutter), the app primarily serves as an inference interface — all major computations and disease classification logic are handled by the deep learning model.

3. IMPLEMENTATION

The implementation phase transforms the designed CNN architecture into a functional deep learning model. It covers dataset loading, model construction, training, tuning, and validation, ensuring that the system effectively learns to classify plant diseases from images. This section explains the practical setup and workflow used to train and optimize the Crop Care model.

3.1. Model Development Environment

The model was implemented using TensorFlow 2.x and the Keras API, providing flexibility and modularity for building neural networks. Training was executed on a GPU-enabled environment to accelerate computation.

Libraries and Tools Used:

- *TensorFlow / Keras:* Core deep learning framework.
- *NumPy, Pandas:* Data handling and preprocessing.
- *Matplotlib:* Visualization of training curves.
- *scikit-learn:* Data splitting and evaluation metrics.

This setup ensured a streamlined workflow for experimentation and model tuning.

3.2. Model Training Process

The training pipeline involved preparing the dataset, defining the CNN, and optimizing the model through iterative learning. The dataset was divided into training and validation subsets (80:20 ratio), with augmentation applied to increase diversity.

Training Workflow:

1. Feed pre-processed batches of images (256×256×3) into the CNN.
2. Compute categorical cross-entropy loss and update weights using Adam optimizer.
3. Monitor training and validation metrics each epoch.
4. Apply callbacks such as EarlyStopping and ModelCheckpoint for optimal performance retention.

Training ran for up to 50 epochs, though early stopping typically halted it earlier upon convergence. The accuracy and loss curves were plotted to visualize performance trends and identify overfitting.

Once the model achieved stable validation accuracy, it was converted to TensorFlow Lite (TFLite) format for mobile deployment. This conversion reduced model size and latency, enabling efficient on-device inference without internet connectivity.

3.3. Evaluation Metrics

The model's performance was primarily assessed using classification accuracy, supported by secondary measures like validation loss and confusion matrix analysis.

- Accuracy: Percentage of correctly predicted samples across all classes.
- Loss: Indicates model's confidence alignment with true labels.
- Confusion Matrix: Helped identify commonly misclassified disease pairs.
- Learning Curves: Illustrated convergence behaviour across epochs.

High validation accuracy and minimal overfitting confirmed the CNN's capability to distinguish diseases effectively, even under variable image conditions.

3.4. Hyperparameter Tuning and Regularization

Optimizing hyperparameters and applying regularization were critical to enhancing model stability.

Key Tuned Parameters:

Parameter	Value	Purpose
Learning Rate	0.001	Controls weight update magnitude
Batch Size	16	Balances speed and gradient stability
Epochs	50 (with early stop)	Defines training cycles
Dropout Rate	0.5	Prevents neuron co-adaptation

Table 1. Parameters

Regularization Techniques:

- *Dropout:* Randomly deactivates neurons (50%) to prevent overfitting.
- *Data Augmentation:* Introduces synthetic variations for improved generalization.
- *Early Stopping:* Automatically halts training once validation performance stabilizes.

Together, these strategies helped the CNN achieve a strong balance between accuracy and computational efficiency — suitable for real-time, mobile-based plant disease detection.

4. RESULTS AND CONLUSION

This section presents the outcomes of the deep learning model after training and testing. It summarizes the model's performance in terms of accuracy, validation results, and prediction capability, followed by a discussion of insights and future improvements. Visual results such as confusion matrices, accuracy/loss plots, and sample predictions support the findings presented here.

4.1. Experimental Results

The model was trained for up to 50 epochs with early stopping applied to prevent overfitting. Throughout training, both training and validation accuracy showed a steady upward trend, while the corresponding losses decreased, confirming effective learning and good generalization.

- Final Training Accuracy: ~98%
- Validation Accuracy: ~95% (average across 15 classes)
- Loss: Converged smoothly with minimal fluctuations

The confusion matrix indicated strong performance across most classes, with only minor misclassifications in visually similar disease types (e.g., early and late blight in potato).

When tested on unseen real-world leaf samples, the model consistently provided correct classifications with high confidence (>90% in most cases).

The TensorFlow Lite (TFLite) version of the model maintained near-identical performance with negligible accuracy drops (<1%), confirming the success of model compression and mobile optimization.

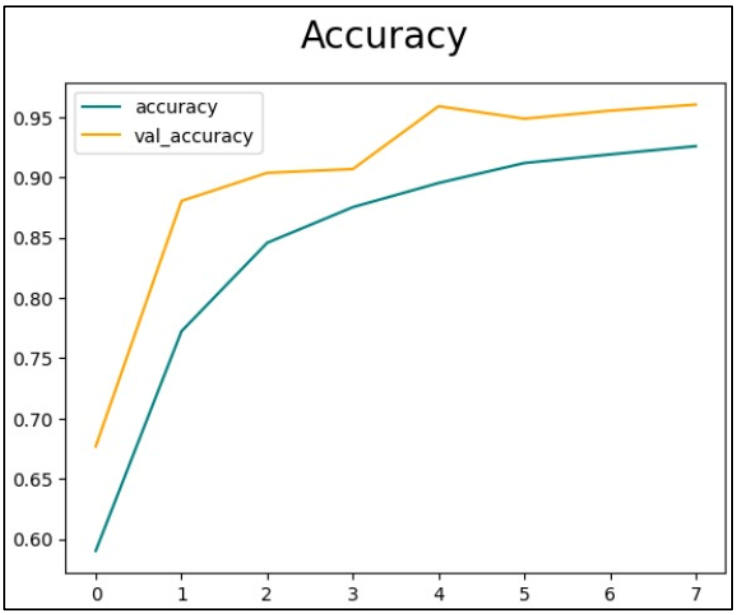


Fig 2. Model Accuracy

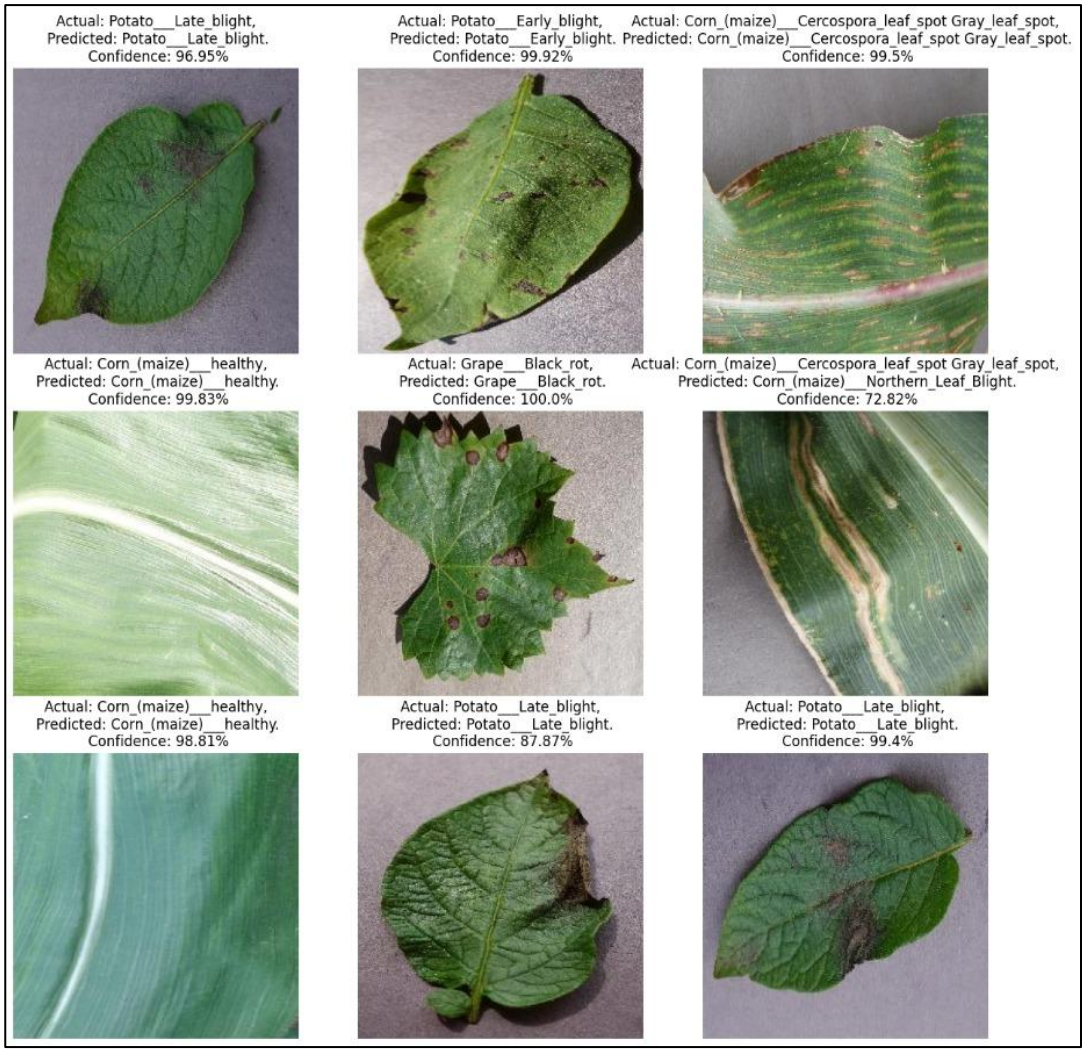


Fig 3. Actual vs Predicted

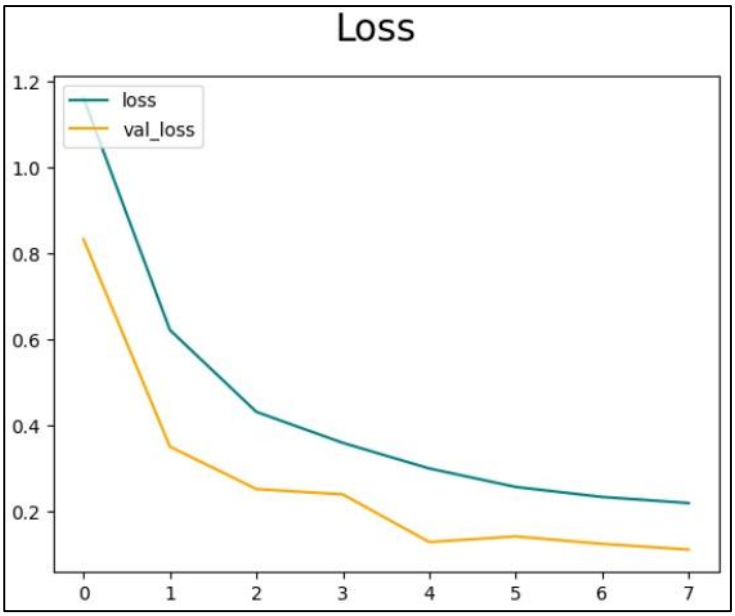


Fig 4. Loss

4.2. Model Performance

The trained CNN demonstrated its ability to extract complex spatial features from leaf images, effectively differentiating disease textures and colour variations. The increasing filter depth (32→128) across layers helped capture both low-level and high-level features, contributing to its strong classification accuracy.

The combination of data augmentation and dropout regularization proved highly effective in reducing overfitting, while the Adam optimizer ensured smooth and adaptive convergence.

On-device testing through the TFLite model showed that predictions were generated in under two seconds per image on average, enabling real-time plant disease detection. The lightweight architecture and quantization further allowed efficient inference without the need for internet connectivity, demonstrating the project's potential for field-level agricultural use.

4.3. Conclusion and Future Work

The Crop Care system successfully demonstrates the integration of deep learning into agriculture through automated plant disease detection. By leveraging a custom CNN architecture, the project achieves high classification accuracy while maintaining computational efficiency suitable for mobile deployment.

This work highlights how deep learning-based computer vision can empower farmers with accessible, instant, and reliable disease diagnosis — reducing dependence on manual inspection and improving timely crop management decisions.

Future Enhancements:

- Expanding the dataset to include more crops and regional disease variations.
- Exploring transfer learning with pre-trained models for faster convergence.
- Implementing explainability techniques such as Grad-CAM to visualize decision regions.
- Extending the mobile app to provide treatment suggestions or connect to agricultural databases.

In conclusion, this project represents a complete deep learning pipeline — from data preprocessing and CNN training to real-world deployment — underscoring the transformative potential of AI in precision agriculture.